

Project 1 — readme

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Before you submit. Fill in both sections, delete these instructions, then export this document to PDF and save it as `readme.pdf` inside your `WebServer` folder. In Word: File → Save As, and choose PDF. Submit the PDF, not this `.docx` file. **Two pages maximum. The TA stops reading at the end of page two.**

1. AI use (~1 page)

AI assistance is allowed. The condition is that you understand every line you submit and can explain it if asked.

If you used no AI tools at all, delete the questions below and write this one line:

I did not use AI tools on this project.

Otherwise, answer all of them.

Which two or three prompts did the most work?

Two prompts that I frequently fell back on was either an explanation of the required function without coding so I could at least make an attempt to implement the function, and the second prompt was syntax and code structure: I am unfamiliar with network programming and the TCP stack, so I relied on prompting and googling to implement to ensure code was in the proper order. The final prompt I used most frequently was further testing and verification that my code worked after following the “Test your work” and “Check these too:” sections.

What did the AI get wrong, leave incomplete, or state confidently but incorrectly?

AI would frequently incorrectly assert that my indentation was wrong. It knew Python has strict indentation but pasting code sometimes stripped indents which would set off the AI. It would also output verbose answers that would go over the scope I wanted to be in, whether that be implementing functions I did not ask for, or providing too much information up front.

What did you do instead?

If anything, I was confident about, it would be correct indentation. Its concerns were promptly ignored or told to disregard seemingly incorrect indentation. As for out of scope implementations, I would ‘hand pick’ the information that was actually helpful and productive to what I wanted to do, or push back and set boundary of what is to be completed.

2. Design (~1 page)

A short paragraph for each question, about one page in total. Write about your code, not about HTTP in general.

How does your server know a request has ended?

My server knows when a request has ended in the `recv_request_head(conn)` function. Given that TCP network connections knows no concept of requests and is just a byte stream, I had to implement a way for it to recognise when a request finished. This was done with a while loop that looped forever until it spotted the `"\r\n\r\n"` in the buffer. This was indicated to in the instructions of Task #3: "Continue calling `recv()` and appending to a buffer until the blank line `\r\n\r\n` that terminates the header block appears; only then is a complete request in hand." That is how I know that the server knows a request ended.

Why does your client not simply read until the connection closes?

This is because as per the spec: "Connections are persistent." The server wants to keep the same connection open after sending a response so that the client can send another. It's also why the implementation uses length of data to exit the loop, because TCP sends a byte stream with no concept of being 'done', so I had to implement a way to exit under a certain condition.

How does your thread pool bound concurrency — and what did you observe when you tested it?

With the default 8 threads created in main from the command line arguments, and are all immediately put into the `work_queue`. It is ultimately `handle_connection` that sends these workers to do something. So threads are bounded by the number of workers available at start up rather than dynamically making more up to a certain limit. Any more requests will have to wait in the queue until at most 5 seconds as per the `conn.settimeout(5)`. Because the server is pretty fast at handling connections, I had to intentionally slow it down with `time.sleep(1)`, and using a set number of threads and a lot of connections, was able to see discrete simultaneous completions of the number of threads I had set.

How this document is marked (not counted in the page limit)

5 marks for each section, awarded 0, 3 or 5.

	0	3	5
AI use	absent	present but generic — could describe any project	specific: real prompts, a real mistake, what you did about it
Design	absent, or plainly not about your own code	broadly right, thin	correct and specific about your own code, not about HTTP in general

This document is worth 10 of the project's 100 points. If it is missing from your submission, it scores 0 on this item and affects nothing else — the automated marks for the five tasks are unaffected.

You may also be picked at random for a short oral check-in and asked to walk through your own code. The design questions above are the questions you will be asked, so answering them honestly here is the preparation.